"I, uh, forgot the root password"

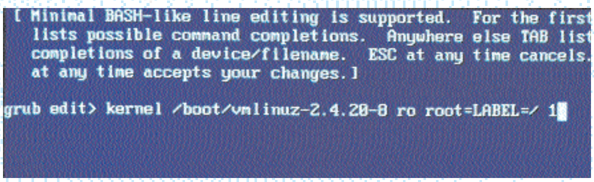
Let's look at recovering the root password from the boot loader. If you're using GRUB, then, as GRUB loads up, highlight the Red Hat Linux entry on the GRUB menu, and then press [E] to edit the boot configuration. Locate the following line, something that looks like this:

kernel /boot/vmlinuz-2.4.20-0.70 root=LABEL=/ hdc=ide-scsi

Type the number *'1'* at the end. Doing so boots the PC into run level 1-single user mode, where you're automatically logged in as root. This done, type *'passwd'* at the prompt. You can enter a new password here.

If you use LILO, try this. At the LILO boot prompt, type *'linux single'* to get to single user mode. Of course, you may have entered a boot password, and forgotten that as well, which effectively prevents you from adding in any arguments to the boot loader whilst booting.

For wretched times like that, get hold of the Red Hat 9 install



Edit the GRUB entry to boot into single-user mode

CDs. Next, boot using the first CD—the bootable CD—and type *'linux rescue'* at the boot prompt. The rescue program will run through a couple of the initial install steps, such as setting up the mouse, keyboard and network interfaces.

You don't need to set up the network interface: run straight through the program to the point where you can choose to mount the hard drive with read-write permissions. Click on Continue, and get into the command prompt. Type *'chroot /mnt/sysimage'*. You now get into the root environment (/) with the right permissions. Now, reset the password by typing *'passwd'*, then type *'exit'* to get out of the chroot-ed environment, and *'exit'* again to reboot. That's it, password recovered.

Remember that since it's so easy to recover a boot password, you need to physically secure your machine if you don't want someone else using your computer as root.

"I can't boot into Linux"

If you have your boot diskette, you should be able to boot into Linux, and reinstall the boot loader. If not, no problem—Linux is quite forgiving, and with just a little effort, you can get back to GRUB-ing or LILO-ing your boots.

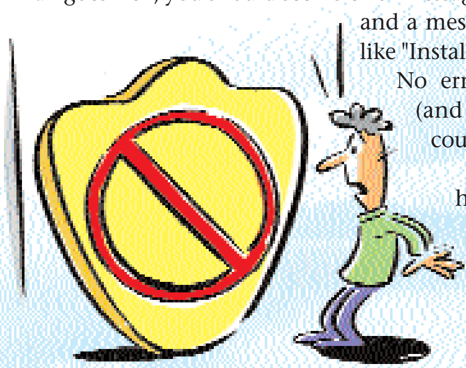
For GRUB users, boot into Linux using the first Red Hat 9 install CD, and once on the prompt, chroot to the */mnt/sysimage* directory. Now that you're in, run this command to get GRUB working:

grub-install /dev/hda

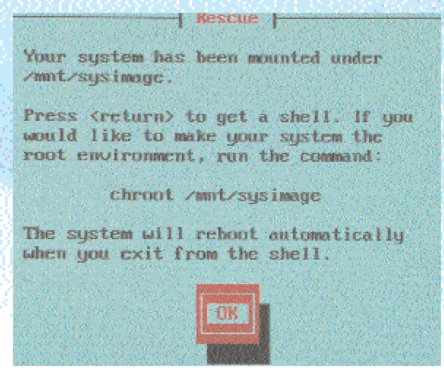
This example assumes that you're loading GRUB in */dev/hda*, the MBR of the first IDE hard drive. To install GRUB in the boot sector of the Linux partition (located on, say, */dev/hda5*), use:

grub-install /dev/hda5

If all goes well, you should see no error messages popping up, and a message that looks like "Installation finished. No error reported..." (and so on, for a couple of lines.)



LILO users have it even easier. All you need to do is run 'lilo' from the command line after booting into Linux using the install CD, and chroot-ing into the correct directory. Of course, in both cases, you would need to have copies of the config files (the grub.conf and the lilo.conf files respectively, located within */etc*) in the right places, and the associated files in the */boot* partition.



Boot into Linux using the Red Hat CDs, the rescue-image way

You may also be unable to boot into Linux because of a kernel panic, which occurs when the kernel cannot be booted into. This could happen because of the installation of a newly , (and badly), compiled kernel. Another problem may be a system service that refuses to load, hence preventing a boot. In such a case, boot into single user mode from the boot loader, and start the ntsysv utility; locate the service and disable it from the bootup list. Other problems could include hard drive corruption errors.